CheckIn – Progress Report

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Summary

CheckIn now runs end to end locally, and this round adds real attendance logic, cleaner teacher pages and a rebuilt admin area. The code is on `feature/frontend-backend-integration` and still needs to be pushed.

- **Start Attendance** gives each session a unique 6-character code that lasts 5 minutes. Students who don't check in are marked absent automatically.
- **Flagged check-ins** catch location and network problems instead of rejecting students, and teachers see the reason.
- **Attendance Records** and **Reports & Analytics** show only real data, start at 0%, and say "Not assigned" until an admin assigns modules.
- **Admin** has a new overview and a page per programme listing its students and teachers, with activate and deactivate buttons.
- **Sign-up** now lists all 11 programmes and 8 departments.

Getting it running

The app needs a working Supabase connection, and **college Wi-Fi blocks it**: ports 5432 and 6543 both failed from the `10.2.x.x` network. Use a phone hotspot when testing on campus.

1. Pull the branch: `git checkout -b feature/frontend-backend-integration origin/feature/frontend-backend-integration`
2. In `backend`, run `npm install`, then create `backend/.env` (ask the team lead for the values, never commit it).
3. In `.env`, both URLs must use the pooler host with `postgres.<project-ref>` as the username, not just `postgres`. Otherwise you get "no tenant identifier provided".
4. Keep `PORT=5000`, because the frontend calls `localhost:5000`.
5. Optional new settings: `CHECKIN_WINDOW_MINUTES=5` and `NETWORK_GRACE_SECONDS=120`.
6. Run `npx prisma generate`, then `node seed.js` (safe to re-run, it skips existing rows), then `npm start`.
7. In `frontend`, run `npm install`, then `npm run dev` (there is no `npm start` script). Open `http://localhost:5173`.

Two things to avoid: don't run `npm audit fix --force`, and don't upgrade Prisma to 8.x. Both can break the shared setup.

Teacher side

Teachers only ever see modules an admin has assigned to them, and every number comes from real check-ins.

Start Attendance (now its own page, `StartAttendance.jsx`)

- The Section field is gone. The module dropdown lists only assigned modules, or says "No modules assigned. Contact admin."
- Generating a code needs the teacher's location. The code shows with a 5:00 countdown that turns red in the last minute.
- Each code is new and never reused, and it skips look-alike characters (0/O, 1/I).
- When time runs out, everyone enrolled who didn't check in is saved as Absent. The backend does this every minute, so it works even if the teacher closes the page.
- Live counts show Present, Absent and Flagged, with each flagged student's reason.

When a check-in is flagged instead of marked present:

Reason shown	When it happens
Location not shared	Student denied location permission
Browser could not get location	The browser's location API failed
Weak GPS accuracy	Accuracy worse than 50 m, or the GPS error margin could put them inside the room
Outside classroom area	Clearly beyond the 10 m geofence
Sent late (poor internet)	Check-in held back by bad internet, arriving up to 2 minutes after the code expired

Attendance Records (`Records.jsx`): a module list, then a student table (practical, theory, total and a flagged count), then one student date by date with Present, Absent or Flagged. Hovering a flag shows the reason. Both pages can download a CSV.

Reports & Analytics (`Reports.jsx`): weekly bars, overall rings and module performance bars from real data only, refreshing every 30 seconds. With no modules it shows "Not assigned" at 0%.

The teacher search box in the top bar was removed.

Admin side

The admin area now starts with a cleaner overview and lets the admin manage people per programme.

- **Overview** (`AdminOverview.jsx`): Total Students, Total Tutors and Total Programs as three large cards on one row, with Students view and Tutors view side by side beneath. Both dropdowns list all 11 programmes.
- **Programme page** (`ProgramView.jsx`): picking a programme opens a page with total and active counts for students and teachers. Tabs switch between a student table (name, ID, attendance, status) and a teacher table (staff ID, name, department, status). Each row has Activate or Deactivate with a confirm step, plus a search box.
- Each programme page lists only that programme's people, filtered both in the backend and the frontend.

- **Assigning a module** stays under Program / Module: pick programme, year, semester, open a module, and type the teacher's staff ID into "Assign Tutor (Staff ID)". This is what makes the module appear for that teacher.
- `Admin.jsx` was rewritten in a readable layout and now only holds the year-wise student list, programmes, semesters and the module form.

Sign-up changes

Students and tutors now pick from the college's full lists in `Register.jsx`, instead of the two programmes and one department it had before.

- **Students** choose a Programme.
- **Tutors** choose a Department and a Programme.
- **Programmes (11):** B.E. Software Engineering, B.E. Information Technology, B.E. Electronics and Communication, B.E. Civil Engineering, B.E. Electrical Engineering, Bachelor of Architecture, B.E. Engineering Geology, B.E. Instrumentation and Control Engineering, B.E. Water Resource Engineering, Master in Construction Management, Master of Science in Engineering (by Research).
- **Departments (8):** Architecture, Civil & Environmental Engineering, Computing Technologies, Electrical & Electronics Engineering, Mechanical Engineering, Science & Humanities, Dean of Academic Affairs, Dean of Student Affairs.

The programme names are stored exactly as written, so the admin list in `admin/programmes.js` must stay identical to the sign-up list. Otherwise students won't appear on their programme page.

Backend changes (`server.js`)

Every existing route is still there; the file is now split into numbered sections (Config, Helpers, Auth, Student, Teacher, Evidence, Admin, Management) with comments.

- `POST /api/sessions`: 5-minute codes (was 10), unique codes, requires the teacher's location, returns `expiresAt`.
- `POST /api/attendance/checkin`: the flag rules in the table above. It also accepts `locationError` and `offline` from the app.
- `GET /api/sessions/:id/live`: adds each flagged student's reason, keeps counts correct after closing, and closes the session once time is up.
- `POST /api/sessions/:id/end`: closes the code now and marks the rest absent.
- A timer runs every minute to close expired sessions and record absences.
- `GET /api/teacher/records/...`: adds a flagged count per student and Present/Absent/Flagged with reasons per date.
- `GET /api/admin/users`: new `?programme=` filter, so programme pages only list their own people.

No database schema change was needed. It uses the existing `Absent` and `Flagged` statuses and the `flag_reason` column.

Files added and changed

File	New or changed	What it does
`backend/server.js`	Changed	Restructured; check-in rules, auto-absent, flags, programme filter
`frontend/src/App.jsx`	Changed	Wires in the new pages; search box removed
`pages/public/Register.jsx`	Changed	Full programme and department lists
`pages/teacher/StartAttendance.jsx` + `start.css`	New	Start Attendance page with timer
`pages/teacher/Records.jsx` + `records.css`	Changed + new	Module list, student table, per-date view with flags
`pages/teacher/Reports.jsx` + `reports.css`	Changed	Real-data charts, "Not assigned" state
`pages/admin/AdminOverview.jsx` + `adminoverview.css`	New	Admin overview
`pages/admin/ProgramView.jsx` + `programview.css`	New	Per-programme students and teachers
`pages/admin/programmes.js`	New	Shared programme list
`pages/admin/Admin.jsx`	Changed	Rewritten for readability

Next steps and open questions

- ☐ UgayNobu: add `gantay11` as a collaborator on GitHub so this work can be pushed
- ☐ Decide which branch is the main working branch; `dev` is the default but empty
- ☐ Admin: assign modules to teachers so Start Attendance, Records and Reports have data
- ☐ Set a programme for older accounts (seeded users, Test Tutor 01/02); without one they don't appear on any programme page
- ☐ Student check-in page: send `locationError` and `offline` so those flag reasons get used
- ☐ Teacher review screen for flagged check-ins (the accept and reject routes already exist)
- ☐ Replace the placeholder `JWT_SECRET` with one shared secret for everyone
- ☐ Stop showing raw database errors on the login page before submission
- ☐ Plan an upgrade of multer 1.x, which has known vulnerabilities

Open question: should flagged check-ins count toward attendance %? Right now they don't count until a teacher accepts them.